

Felicity tech tree: Gallium Arsenide photovoltaics

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Abstract

Math behind the photovoltaic power generation in the Felicity tech tree.

References an earlier "[Ion beam printing](#)" [document by me](#), as well as the paper "[Overview of the Current State of Gallium Arsenide-Based Solar Cells](#)" by Nikola Papež, Rashid Dallaev, Stefan Talu and Jaroslav Kaštyl.

Contents

1	Characteristics of GaAs photovoltaics	2
2	Construction of concentrated solar arrays	3
3	Concentrated solar array model	5

Reference markings:

[1] - "[Ion beam printing](#)"

[2] - "[Overview of the Current State of Gallium Arsenide-Based Solar Cells](#)"

[3] - [wikipedia page for "Gallium Arsenide"](#)

[4] - "[Felicity tech tree: low temperature electrolysis of mixed metal oxides](#)"

[5] - Engineering Toolbox table of emissivities for various materials

(https://www.engineeringtoolbox.com/emissivity-coefficients-d_447.html)

1 Characteristics of GaAs photovoltaics

GaAs photovoltaics are made 10 μ m thick [2], and achieve an efficiency of 40%.

Given that Gallium Arsenide has a density of 5.3176 g/cm³ [3], the mass of material that goes into creating the panels would be 53.176g/m².

$$\frac{5.3176g/cm^3}{10\mu m} = 53.176g/m^2$$

Given that molar mass of Gallium Arsenide is 144.645 g/mol [3], the amount of material needed to create the panes is around 0.368mol of GaAs per square meter. This means 0.368mol of Gallium and 0.368mol of arsenic, as GaAs contains both in equal parts, giving a total of 0.735 mol/m² counting individual elements. For the individual elements:

- Gallium : 69.72 g/mol, results required 0.36763mol/m², 25.6312g/m²
- Arsenic : 74.9216 g/mol, results required 0.36763mol/m², 27.5435g/m²

$$\frac{53.176g/m^2}{144.645g/mol} = 0.3676310968mol/m^2 \quad 2 \cdot 0.3676310968mol/m^2 = 0.7352621936mol/m^2$$

$$(0.3676310968mol/m^2) \cdot (69.72g/mol) = 25.63124007g/m^2(Ga)$$

$$(0.3676310968mol/m^2) \cdot (74.9216g/mol) = 27.54350998g/m^2(As)$$

The printer discussed in [3] is capable of depositing 5 micromol per second, and consumes 202.62MJ/mol. As such the panels would take around 147ks/m² and 149MJ/m² to print.

$$\frac{0.7352621936mol/m^2}{5E-6mol/s} = 147052.4387s/m^2$$

$$0.7352621936mol/m^2 \cdot 202.62MJ/mol = 148978825.7J/m^2$$

All taken together:

Solar conversion efficiency	40%
mass	25.6312g/m ² Gallium 27.5435g/m ² Arsenic total 53.176g/m ²
material amount (single atoms)	0.36763mol/m ² Gallium, 0.36763mol/m ² Arsenic, total 0.73526mol/m ²
time to print	147.052ks/m ²
energy to print	148.979MJ/m ²

2 Construction of concentrated solar arrays

GaAs photovoltaics can handle sunlight at intensities of 246 to 302 suns [2], as such, ignoring absorbtion for now, the area of the collector can be that many times smaller than the mirror area.

As a basic standard design I will assume a concentrated solar array where the collector is a disk 5m in radius, and the mirror as seen from the front is 50m in radius.

Covering a 5m-radius collector in GaAs photovoltaics would require printing a total area of 15.708m^2 . This results in the following resource costs:

- Gallium: 5.775mol, 0.402615kg
- Arsenic: 5.775mol, 0.4326533kg
- Energy: 2340.162132 MJ
- Time: 2.340162132 Ms ($\approx 27 \text{ d} + 2 \text{ h} + 2 \text{ min} + 42.132 \text{ s}$)

$$M_{Ga} = 25.6312\text{g}/\text{m}^2 \cdot 15.708\text{m}^2 = 0.4026148896\text{kg}(Ga)$$

$$N_{Ga} = 0.36763\text{mol}/\text{m}^2 \cdot 15.708\text{m}^2 = 5.77473204\text{mol}(Ga)$$

$$M_{As} = 27.5435\text{g}/\text{m}^2 \cdot 15.708\text{m}^2 = 0.432653298\text{g}(As)$$

$$N_{As} = 0.36763\text{mol}/\text{m}^2 \cdot 15.708\text{m}^2 = 5.77473204\text{mol}(As)$$

$$E = 148.979\text{MJ}/\text{m}^2 \cdot 15.708\text{m}^2 = 2340.162132\text{MJ}$$

$$T = 148.979\text{ks}/\text{m}^2 \cdot 15.708\text{m}^2 = 2340162.132\text{s}$$

The skeleton of the concentrated solar array has a volume of 78.7869m^3 , measured empirically by rendering a 3d model (code at the end) and measuring the volume in Orca slicer. The assumed thickness for both the mirror, supports, and the collector was 1cm.

Assuming metal density of $7750\text{kg}/\text{m}^3$, the skeleton would mass 610598.475 kg.

Given that the LTE process [4] can produce metal at a cost of 14.7625MJ/kg, and rate of 1.008kg/h, that amount of metal would take 9014GJ and 2.18071Gs to produce.

(2.18071Gs $\approx 25239 \text{ d} + 16 \text{ h} + 27 \text{ min} + 19 \text{ s}$)

$$78.7869\text{m}^3 \cdot 7750\text{kg}/\text{m}^3 = 610598.475\text{kg}$$

$$610598.475\text{kg} \cdot 14.7625\text{MJ}/\text{kg} = 9.013959987\text{TJ}$$

$$\frac{610598.475\text{kg}}{1.008\text{kg}/\text{h}} = 2180708839\text{s}$$

The concentrated solar array is collecting light with a mirror 50m in radius, but the collector which is 5m in radius is in front of the mirror, blocking some of the light. The mirror is also only 93% reflective since smooth polished steel absorbs 7% of light [5]. The photovoltaics have an efficiency of 40%.

$$(\pi(50m)^2 - \pi(5m)^2) \cdot 0.93 \cdot 0.4 = 7775.441818m^2 \cdot 0.93 \cdot 0.4 = 2892.464356m^2$$

The array is equivalent to a 100% efficient collector with an area of $2892.464356m^2$.

At earth's distance, sunlight intensity is $1361W/m^2$, which means the array would produce 3.9366MW of power.

$$2892.464356m^2 \cdot 1361W/m^2 = 3936643.989W$$

This can be generalized for largely any level of insolation, for example, the asteroid belt, at the distance of 2.7AU, has an insolation of $186.69W/m^2$, so the array would produce 540kW of power.

$$1361W/m^2 \cdot \left(\frac{1AU}{2.7AU}\right)^2 \cdot 2892.464356m^2 = 186.6941015W/m^2 \cdot 2892.464356m^2 = 540006.0341W$$

And at Saturn's distance, 9.5826 AU, the insolation is only $14.8215W/m^2$ so the array would produce 42.87kW of power.

$$1361W/m^2 \cdot \left(\frac{1AU}{9.5826AU}\right)^2 \cdot 2892.464356m^2 = 14.82147429W/m^2 \cdot 2892.464356m^2 = 42870.58609W$$

Cost of the array, all taken together:

- 610598.475 kg of mixed metal alloy in the mirror and support skeleton
- 0.402615kg (5.775mol) of Gallium in the photovoltaic cells
- 0.4326533kg (5.775mol) of Arsenic in the photovoltaic cells
- 2340.162132 MJ energy spent on beam printing the photovoltaic cells
- 2.340162132 Ms total beam print time for the photovoltaic cells

Ofcourse additional energy must be spent producing the materials if they are not available:

- 9.013959987 TJ energy spent on electrolysis for the mixed metal alloy
- 2 180 708 839 s total electrolysis time for the mixed metal alloy
- todo: Germanium and Arsenic production costs, that will have to wait for a future document

3 Concentrated solar array model

Here is the OpenSCAD code for the 3d model that was used to calculate the volume of the concentrated solar array.

```
module_parabolic_mirror(focal_length=20,mirror_length=80,thickness=1,detail=16){
    //Parameters
    focal_length; //focal_length(mm)
    z_max; //mirror_depth(mm)
    segments; //detail; //increase for smoother curve
    thickness; //rim_thickness(optional)

    //build_2D_profile_points(x,z)
    pts=[for(i=[0:segments])let(z_i=z_max*i/segments,x_i=2*sqrt(f*z_i))x_i,z_i];
    //optionally close inner vertex and add rim thickness
    profile=concat([0,0],pts,[0,z_max+thickness]);

    difference(){
        color("#888"){rotate_extrude($fn=segments)polygon(points=profile);}
        translate([0,0,thickness]){
            color("#CCC"){rotate_extrude($fn=segments)polygon(points=profile);}
        }
    }

    //100,80,180
    //50,80,130
    //50,50,100
    //so the radius of the mirror as seen from head on is just the focal length plus mirror length

    module_concentrated_solar_array(array_radius,collector_radius,thickness=1,detail=16){
        //
        focus=array_radius*(3/4);
        depth=array_radius*(1.33/4);
        //1.33 derived empirically to achieve correct radius
        //
        focus=array_radius;
        depth=array_radius/4;
        //this new way of calculating the focus and depth also results in correct size
        //but this time it feels way too elegant to be a coincidence
        //I don't know the math behind why it should work out this way
        parabolic_mirror(focus,depth,thickness,detail);
        color("#888"){
            translate([0,0,focus]){cylinder(r=collector_radius,h=thickness);}
            for(i=[0:1:8]){
                rotate([0,0,i*45]){
                    hull(){
                        translate([array_radius-thickness,0,depth]){sphere(thickness/2);}
                        translate([collector_radius,0,focus+thickness/2]){sphere(thickness/2);}
                    }
                }
            }
        }
    }

    module_mobile_concentrated_solar_array(array_radius,collector_radius,thickness=1){
        concentrated_solar_array(array_radius,collector_radius,thickness);
        translate([0,0,-array_radius/2]){cylinder(r=0.5,h=2*array_radius);}
        translate([0,0,-array_radius/2]){sphere(r=3);}
        translate([0,0,1.5*array_radius]){sphere(r=3);}
    }

    rotate([0,90,0]){
        concentrated_solar_array(50,5,0.01,32);
    }

    $vpr=[105,0,35];
    $vpd=325;
    _____
```